

CRYSTAL PM HOW-TO GUIDE

Set up the Crystal Meet dashboard

One small always-on Raspberry Pi on the office network runs the dashboard that every room reports to. One command installs it; the same command updates it.

Do this once, before the first room. Plan about 30 minutes, most of it waiting for the Pi to download and build. The dashboard is where you create a token for each room, see which rooms are online, and watch their meetings. Rooms keep joining calls even when the dashboard is off; they only show Offline until it is back.

STEP 1

Prepare the Pi

Pi OS Lite 64-bit, hostname crystal-meet, wired.

STEP 2

Fix its address

A DHCP reservation so the address never changes.

STEP 3

Install

One command: Docker, the dashboard, backups.

STEP 4

Sign in

The printed password, then a new one.

STEP 5

Rooms

A token per room and the address in each config.

STEP 6

Keep it running

Backups, updates, logs.

BEFORE YOU BEGIN

Hardware. A Raspberry Pi 4 or Pi 5 with 4 GB or more, its official power supply, an Ethernet cable to the office network, and a 32 GB microSD card. A USB SSD instead of the card is kinder to a database but not required. No screen or keyboard: you work over SSH from your computer.

Access. A computer with Raspberry Pi Imager, SSH, and the login to your router (UniFi) to reserve an address.

What this is not. It is not one of the room devices. Keep the dashboard on its own Pi: a room Pi is busy during calls and the first thing to be unplugged or reinstalled.

1 Prepare the Pi

- **Flash the card.** In [Raspberry Pi Imager](#) choose your Pi model, then [Raspberry Pi OS \(other\)](#) › [Raspberry Pi OS Lite \(64-bit\)](#), and the card or SSD. Lite has no desktop; the dashboard needs none.
- **Pre-fill the settings.** When Imager offers customisation, set the hostname to `crystal-meet`, a username (for example `pi`) with a password you keep, your time zone, and turn on SSH with password authentication. Skip Wi-Fi: use the cable. Write the card.
- **Boot it.** Put the card in, connect Ethernet and power, wait two minutes.
- **Sign in from your computer.** Open a terminal and type `ssh pi@crystal-meet.local` (your username instead of pi). If the name does not resolve, find the Pi's address in your router's client list and use that.

You should now see a shell prompt on the Pi.

2 Give it a fixed address

- **Find the address.** On the Pi type `hostname -I`; the first number, for example `10.0.0.50`, is its address.
- **Reserve it.** In the UniFi Network app open [Client Devices](#), pick `crystal-meet`, open its settings and turn on [Fixed IP Address](#) with that address. Any other router has the same thing under DHCP reservations.
- **Write it down.** This address goes into every room's config in step 5.

You should now see the Pi listed in the router with a fixed address.

3 Install the dashboard

- **Get the code.** On the Pi type `sudo apt install -y git`, then `git clone https://github.com/ben-abeo/croom.to.git && cd croom.to`.
- **Run the installer** with the email address of the first admin, which is you: `sudo bash installer/install-dashboard.sh --admin-email you@crystalpm.com`. It installs Docker, copies the code to `/opt/croom-dashboard`, writes the secrets to a private file, builds the dashboard image, starts it, and sets up nightly backups. The build takes about ten minutes on a Pi; the screen stays busy.
- **Read the last lines.** They show the dashboard's address, the admin email, and a password shown this once. Copy the password into your password manager now.

You should now see "The Crystal Meet dashboard is running" and the address lines.

GOOD TO KNOW

The secrets live only in `/opt/croom-dashboard/deploy/dashboard/.env` on the Pi, readable by root. The installer never rewrites that file, so running it again later is safe: it only updates the code and rebuilds.

4 Sign in and change the password

- **Open the dashboard** from any computer on the office network at <http://crystal-meet.local> or <http://10.0.0.50> (your address). Sign in with the admin email and the printed password.
- **Change the password.** Open [Settings](#), fill in [Change password](#) with the printed password and a new one of at least 8 characters, press [Change password](#).

You should now see the Devices page, empty for now, and "Password changed" on Settings.

5 Point the rooms at it

- **One token per room.** On [Provisioning](#), type the room's name and press [Generate Token](#); each token works once. The room setup guide uses it in its step 2.
- **The address in each room config.** In `deploy/rooms/room-N.yaml` set `dashboard.url` to `http://10.0.0.50:3001` (your address, port `3001`) before installing that room.
- **A room that is already installed.** On that room's Pi edit `/etc/croom/config.yaml` with `sudo nano`, set the same `url` under `dashboard`, then type `sudo systemctl restart croom`.

You should now see each room appear on Devices as online within a minute of its install or restart.

6 Keep it running

- **Backups happen by themselves** every night at 02:30 into `/var/backups/croom-dashboard`, 14 days kept, readable by root only. Take one right now with `sudo systemctl start croom-dashboard-backup.service`. To copy one off the Pi, first make a copy you own: on the Pi, `sudo cp /var/backups/croom-dashboard/croom-dashboard-2026-10-05.sql.gz ~ && sudo chown $USER ~/croom-dashboard-2026-10-05.sql.gz`, then from your computer `scp pi@crystal-meet.local:croom-dashboard-2026-10-05.sql.gz .`
- **Restore one** onto a running dashboard, for example after swapping the card and reinstalling: on the Pi, `cd /opt/croom-dashboard/deploy/dashboard`, then `sudo docker compose stop dashboard`, then `sudo gunzip -c /var/backups/croom-dashboard/croom-dashboard-2026-10-05.sql.gz | sudo docker compose exec -T db psql -v ON_ERROR_STOP=1 -U croom -d croom` (it stops at the first error), then `sudo docker compose start dashboard`.
- **Update** whenever the code changes: `sudo bash /opt/croom-dashboard/installer/install-dashboard.sh`. It pulls, rebuilds and restarts; devices, users and tokens stay.
- **Logs.** `cd /opt/croom-dashboard/deploy/dashboard` then `sudo docker compose logs -f dashboard`. `sudo docker compose ps` shows both parts as Up (healthy). Every compose command needs sudo: the settings file belongs to root.
- **Power.** Pull the plug and put it back: the dashboard comes back on its own within about a minute.

You should now see a dated file in the backup folder the morning after the install.

Troubleshooting

The installer stops while installing Docker

The Pi could not reach the internet or apt is broken. Check `ping -c 3 download.docker.com`, then `sudo apt update`, and run the installer again.

The build fails or the Pi freezes

Not enough memory: the installer refuses a Pi with less than 2 GB and warns below 4 GB. Use a 4 GB Pi, or on a 2 GB one add swap with `sudo dphys-swapfile swapoff`, set `CONF_SWAPSIZE=2048` in `/etc/dphys-swapfile`, `sudo dphys-swapfile setup && sudo dphys-swapfile swapon`, and run the installer again.

The page is blank or does not load

On the Pi,
`cd /opt/croom-dashboard/deploy/dashboard`
 then `sudo docker compose ps` and
`sudo docker compose logs dashboard`.
 "Database initialization failed" means Postgres is still starting; wait a minute. Anything else: run the installer again.

Rooms show Offline

From a room's Pi type `curl http://10.0.0.50:3001/health` (your address). No answer means the address in that room's config is wrong or the dashboard Pi is off. An answer with the room still offline means the room's token was already used: make a new one on Provisioning.

Forgotten admin password

Delete the admin and let the next start recreate it from the private file: `sudo docker compose exec db psql -U croom -d croom -c "DELETE FROM users WHERE email = 'you@crystalpm.com';"` then `sudo docker compose restart dashboard`; the first password is on the `ADMIN_PASSWORD` line of `/opt/croom-dashboard/deploy/dashboard/.env` (`sudo cat`).

crystal-meet.local does not resolve

Some networks and tablets do not resolve `.local` names. Use the fixed address instead; that is also why the room configs use the address, not the name.

GOOD TO KNOW

Everything speaks plain HTTP on the office network; nothing is reachable from the internet unless you forward a port, which this guide never asks you to do.